



$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}[\mathbf{R} \ \mathbf{t}] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}, \quad (u, v) = \left(\frac{\tilde{u}}{\tilde{w}}, \frac{\tilde{v}}{\tilde{w}} \right)$$